

MS-GAN: GAN-based Semantic Segmentation of Multiple Sclerosis Lesions in Brain Magnetic Resonance Imaging

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- Multimodal Encoder-Decoder Generator
- Multimodal Discriminators

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Biomedical Background of Multiple Sclerosis (MS)

- An inflammatory disease of the **central nervous system**.
- Can be detected using the magnetic resonance imaging (**MRI**).
- **No known cures** at present.
- Quantitative analysis of the number and volume of lesions is essential during disease diagnosis and management.

Multiple Sclerosis (MS) Lesion Segmentation

- A fundamental step to facilitate these quantitative analyses above.
- A multi-modal segmentation task by nature (multi-modal MRI data).
- Highly challenging due to the large variation of the MS lesions, in terms of their size, location, shape, and intensity.

Biomedical Background of Multiple Sclerosis (MS)

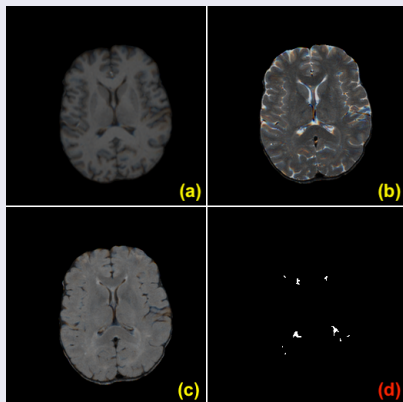
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Multiple Sclerosis (MS) Lesion Segmentation

- **A multi-modal segmentation task** by nature.
- **Highly challenging** due to the large variation of the MS lesions.



- T1-weighted MRI
(anatomical structures).
- T2-weighted MRI
(pathological changes).
- FLAIR MRI
(pathological changes).
- MS lesion annotation.

Discriminators D are optimized to detect the difference between the “real” ground truth and “fake” lesion masks produced by the generator G , against the adversarial learning process of the generator G , where G is trained to output lesions masks that appear as real as possible.

MS-GAN Design:

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- Multimodal Encoder-Decoder Generator
- Multimodal Discriminators

Conditional Generative Adversarial Network (cGAN) [Mirza et al., 2014]

$$L_{\text{cGAN}}(\mathbf{G}, \mathbf{D}) = E_{x,y}[\log D(x, y)] + E_{x,z}[\log(1 - D(x, G(x, z)))]$$

- Generator G tries to minimize this objective against a discriminator D that tries to maximize it, i.e.

$$G^* = \arg \min_G \max_D L_{\text{cGAN}}(G, D)$$

Objective

Conditional Generative Adversarial Network (cGAN) [Mirza et al., 2014]

$$\mathcal{L}_{\text{cGAN}}(\mathbf{G}, \mathbf{D}) = E_{x,y}[\log D(x, y)] + E_{x,z}[\log(1 - D(x, G(x, z)))]$$

pix2pix [Isola et al., 2017]

$$\mathcal{L}_{\text{pix2pix}}(\mathbf{G}, \mathbf{D}) = \mathcal{L}_{\text{cGAN}}(\mathbf{G}, \mathbf{D}) + \lambda \mathcal{L}_{L1}(\mathbf{G})$$

- Add $\mathcal{L}_{L1}(\mathbf{G})$ term to alleviate the hard-to-converge issue of GAN and its variants.
- Learning objective is

$$G^* = \arg \min_G \max_D \mathcal{L}_{\text{pix2pix}}(\mathbf{G}, \mathbf{D})$$

Objective

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MS-GAN

$$\mathcal{L}_{\text{MS-GAN}}(\mathbf{G}, \mathbf{D}) = \mathcal{L}_{\text{cGAN}}(\mathbf{G}, \mathbf{D}) + \lambda [\mathcal{L}_{L1}(\mathbf{G}) + \mathcal{L}_{\text{SemSeg}}(\mathbf{G})]$$

- Maintain the regression-related constraint $\mathcal{L}_{L1}(\mathbf{G})$
- Propose the classification-related constraint $\mathcal{L}_{\text{SemSeg}}(\mathbf{G})$

$$\mathcal{L}_{\text{SemSeg}}(\mathbf{G}) = \mathcal{L}_{\text{CrossEntropy}}(\mathbf{G}) + \mathcal{L}_{\text{Dice}}(\mathbf{G})$$

Key points of design:

- Encoder-decoder architecture.
- Multiple encoding paths for multi-modal MRI data inputted.
- Cross-modality fusion module to encode multi-modal information efficiently.
- **(*Details and Param setting)**

Multimodal Encoder-Decoder Generator

Generator Architecture:

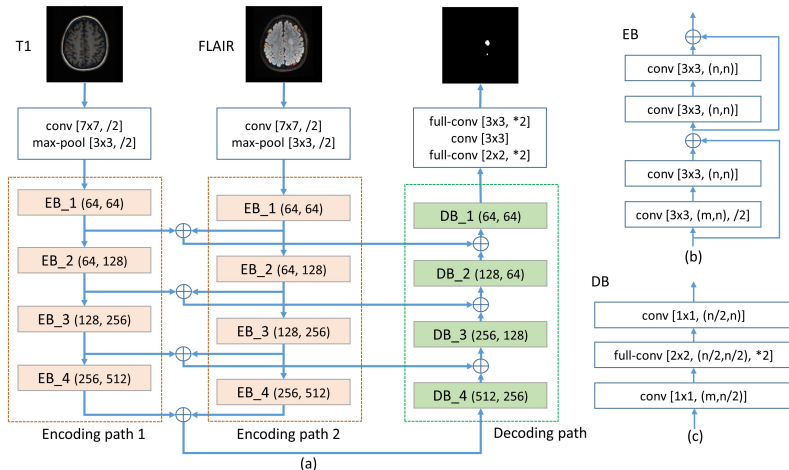


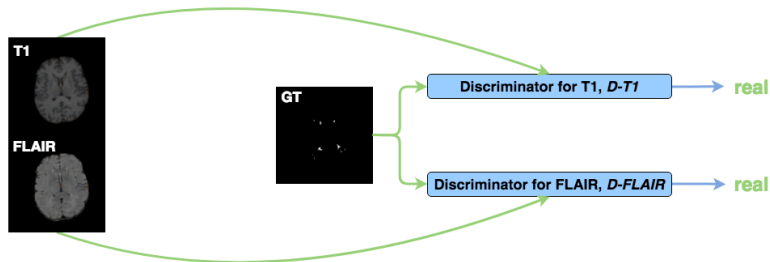
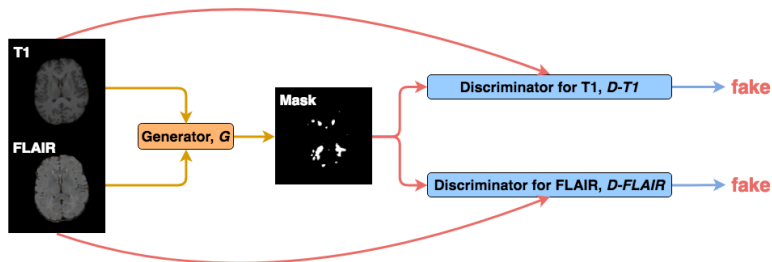
Figure: (a) Proposed multimodal encoder-decoder generator. (b) Encoder block. (c) Decoder block.

- **We adopted PatchGAN as our base model for the discriminators**, which would try to classify whether a given $N \times N$ patch is real or fake. N is set to 50 in our experiments.
- We argue that the visual features used to distinguish patches collected from different input modalities, should differ from each other. Hence, **each modality would construct its own discriminator to focus on the feature extraction**, which is uniquely representative for that specific modality only.

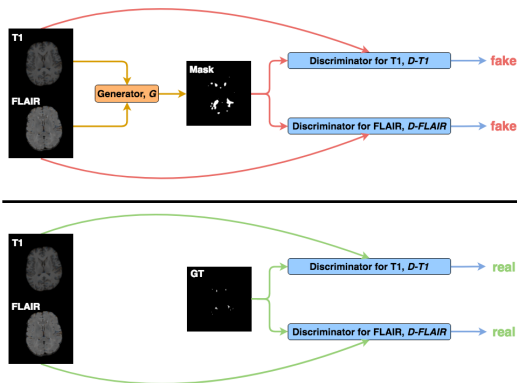
Multimodal Discriminators

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Overall Architecture



Overall Architecture



The training process of MS-GAN is to map from multiple modalities *MODs* to lesion mask *MASK*. Each modality has its own discriminator, D , to distinguish real (ground truth, modality) from fake (mask generated, modality) pairs. The generator, G , is not only targeted at tricking the discriminators, but is also expected to produce semantic segmentation for MS lesions.

Experiments and Results

- Dataset Description
- Ablation Studies
 - Multimodal Generator Architecture Design
 - Classification-related Constraint
 - Over-segmentation and Under-segmentation Issue
- Comparison with the State-of-the-art Approaches

Dataset Description

Dataset Collection

- A total of **69** consecutive patients with relapsing-remitting MS were included in this study (male / female: 33 / 36, mean age: 43.3 ± 11.2). Both baseline and follow-up images were acquired, producing a total of **126** scans. (***Details**)

Dataset Split

- For evaluation, the patients were divided into two groups. The first 9 patients were selected to form the test group, resulting in **20 testing volumes** (2-3 scans per patient). MRI data of the other 60 patients were used as the training data (1-3 scans per patient), resulting in **106 training volumes** (1-3 scans per patient).
- Within the training data, **4-fold cross-validation** is applied.

Dataset Description

Pseudo 3D (P3D) slices

- An axial slice is a grayscale image, we treat one slice as one color channel in the RGB space, and stack three consecutive slices together into one RGB image.
- The resultant color image becomes a pseudo 3D slice capturing the neighbour depth information.

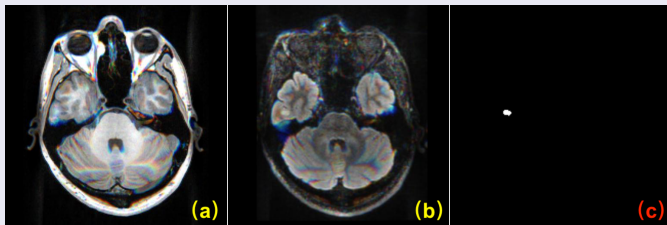


Figure: Pseudo 3D slices encoding depth information as RGB channels. (a) T1-weighted P3D. (b) Flair P3D. (c) MS lesion mask.

Ablation Studies (a): Multimodal Generator Design

	Dice	HD
(a) FLAIR Modality only	0.613	2.07
(b) T1-weighted Modality only	0.191	6.75
(c) Concatenation of Two Modalities	0.623	2.36
(d) Our G with element-wise sum	0.642	2.26
(e) Our G with element-wise multiplication	0.619	2.72
(f) Our G with cross-modality max-pooling	0.554	3.09
(g) Our G with cross-modality convolution [Tseng et al., 2017]	0.628	2.30

Table: Segmentation results evaluating our generator components.

- **Multiple encoding paths design** [from (d) to (g)] is more suitable for multi-modal segmentation problem, such as the MS lesion segmentation task.
- **Element-wise sum** (d) achieves the best result among all cross-modality fusion module design.

Rethink of pix2pix and its regularization term $L_{L1}(G)$

- This design of $L_{pix2pix}(G, D)$ has succeeded in approximating the mapping functions for many image-to-image translation problems.
- However, we found that it could not achieve a similarly accurate result for the semantic segmentation of MS lesions.
- We believe that this is primarily attributable to their design of GAN loss, which **formulates the image-to-image translation problems as regression tasks**, while semantic segmentation is intrinsically a dense classification task performed at the pixel-level.

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(Summary) pix2pix

- Regression-related constraint $L_{L1}(G)$.
- Regression-based image-to-image translation problem.
- For example, the *image colorization problem*, whose output images are expected to be the images of real-valued pixels.

Ablation Studies (b): Classification-related Constraint

Our MS-GAN and its classification-related constraint $L_{SemSeg}(G)$

- We therefore reformulated the problem by incorporating the **classification-related objective term** $L_{SemSeg}(G)$, including $L_{CrossEntropy}$ and L_{Dice} , into the design of our MS-GAN loss.
- $L_{SemSeg}(G)$ should encourage our MS-GAN to formulate this image-to-image translation problem closer to a classification task, since **the nature of semantic segmentation is a pixel-wise classification task**, rather than a regression work.
- This loss improvement alleviates the **hard-to-converge issue** that characterizes MS lesion semantic segmentation. This improvement can be applied to **other classification-based image-to-image translation problems**.

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(Summary) MS-GAN

- Classification-related constraint $L_{SemSeg}(G)$.
- Classification-based image-to-image translation problem.
- For example, the *MS lesion segmentation problem*, whose output images are expected to be the images of discrete-valued pixels.

Ablation Studies (b): Classification-related Constraint

$$\mathbf{L}_{\text{MS-GAN}}(\mathbf{G}, \mathbf{D}) = L_{\text{cGAN}}(G, D) + \lambda[L_{L1}(G) + L_{\text{CrossEntropy}}(G) + L_{\text{Dice}}(G)]$$

	Dice
Generator only	0.642
(1) MS-GAN with $L_{L1}(G)$ removed	0.671
(2) MS-GAN with $L_{\text{CrossEntropy}}(G)$ removed	0.317
(3) MS-GAN with $L_{\text{Dice}}(G)$ removed	0.665
(4) MS-GAN with all constraints preserved	0.672

Table: Segmentation results evaluating various constraints.

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- The significant Dice drop with $L_{\text{CrossEntropy}}(G)$ removed, indicates the appearance of the hard-to-converge issue, resulting from attempting to use GAN to approximate a classification-based mapping function, via treating it as a regression problem.

Under-segmentation and Over-segmentation Issue

Issue Identification

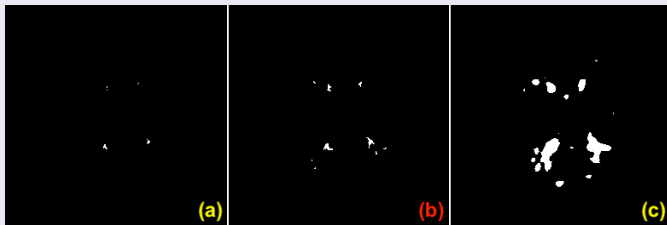
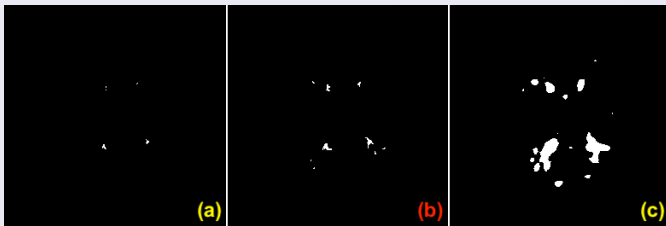


Figure: (a) Under-segmentation. (b) MS lesions ground truth. (c) Over-segmentation.

These issues would occur when the segmentation target is in highly imbalanced situation.

Under-segmentation and Over-segmentation Issue

Issue Identification

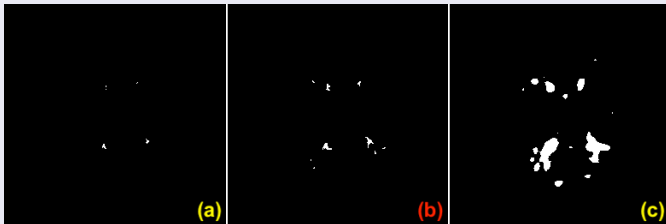


Previous Solutions

- **(Image Processing)** Applying morphological image filtering as an extra post-processing step.
- **(Machine Learning)** Adjusting the cross-entropy weight manually for the loss function.

Under-segmentation and Over-segmentation Issue

Issue Identification



Previous Solutions

- **(Image Processing)** Applying morphological image filtering as an extra post-processing step (highly depends on the segmentation mask produced).
- **(Machine Learning)** Adjusting the cross-entropy weight manually for the loss function (requires a group of empirical pre-experiments).

Under-segmentation and Over-segmentation Issue

MS-GAN Approach

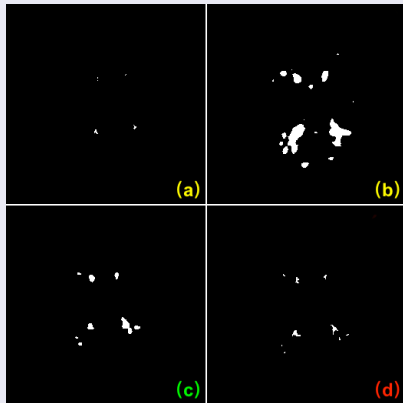


Figure: (c) Mask produced by MS-GAN.

- **(Ability of GAN)**
Approximate complex mapping functions and achieve reasonable results without hand-engineering the loss functions [Isola et al., 2017].
- **(MS-GAN)** Perform MS lesion segmentation without a precise setting for the cross-entropy weight.

Comparison with the State-of-the-art Approaches

	Dice	Recall	Precision	F-score	HD
U-Net [Ronneberger et al., 2015]	0.405	0.376	0.343	0.359	3.99
GCN [Peng et al., 2017]	0.490	0.475	0.459	0.467	3.11
DeepMedic [Kamnitsas et al., 2017]	0.623	0.627	0.498	0.556	1.98
Our MS-GAN	0.672	0.692	0.724	0.708	2.19

Table: Segmentation results, comparing our method with other deep learning approaches.

- We evaluated the segmentation results with the voxel-level Dice coefficient, object-level recall, precision and F-score, and slice-level Hausdorff distance (HD).
- Our method largely outperforms the compared state-of-the-art deep learning approaches. (***Deepmedic is the winner of the ISLES 2015 brain lesion segmentation competition.**)

GAN-based semantic segmentation framework MS-GAN

- Our model consists of one multimodal encoder-decoder generator G and multiple patch-based discriminators D corresponding to the multiple input modalities.
- The multimodal design of our generator effectively integrates the different imaging modalities in the end-to-end learning, and the adversarial training process with the discriminators helps to refine the segmentation.
- We also define a classification-related constraint for the optimization of GAN, which helps to alleviate the hard-to-converge issue for the MS lesion segmentation task.
- Experiments were conducted on a set of 126 cases from clinical studies. The evaluation results show that our method is more effective than the other popular semantic segmentation models and a patch-based 3D CNN model that is customized for brain MRI.

References



Mirza, M. and Osindero, S. (2014)

Conditional Generative Adversarial Nets

arXiv preprint arXiv:1411.1784



Isola, Phillip and Zhu, Jun-Yan and Zhou, Tinghui and Efros, Alexei A. (2017)

Image-To-Image Translation With Conditional Adversarial Networks

The IEEE Conference on Computer Vision and Pattern Recognition (CVPR), July 2017



O. Ronneberger and P. Fischer and T. Brox (2015)

U-Net: convolutional networks for biomedical image segmentation

MICCAI, 234-241 2015



C. Peng and X. Zhang and G. Yu and G. Luo and J. Sun (2017)

Large kernel matters - improve semantic segmentation by global convolutional network

The IEEE Conference on Computer Vision and Pattern Recognition (CVPR), July 2017



K. Kamnitsas and C. Ledig and V. Newcombe and J. Simpson and A. Kane and D. Menon and D. Rueckert and B. Glocker (2017)

Efficient multi-scale 3D CNN with fully connected CRF for accurate brain lesion segmentation

Med. Image Anal., 36 61-78 2017



K. Tseng and Y. Lin and W. Hsu and C. Huang (2017)

Joint sequence learning and cross-modality convolution for 3D biomedical segmentation

The IEEE Conference on Computer Vision and Pattern Recognition (CVPR), July 2017

The End